
Probe-Budget Redundancy in Projected Hessian Learning: Subsampled Hutchinson Estimators Preserve Accuracy at Fivefold Lower Probe Cost

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Abstract

Projected Hessian Learning (PHL) supervises the curvature of machine-learning interatomic potentials through stochastic Hutchinson estimators, avoiding the quadratic cost of explicit Hessian matrices by probing curvature with Hessian-vector products (HVPs). The reference PHL protocol draws one probe vector *per molecule per minibatch*, so a minibatch of B systems costs B HVP evaluations per step. We show that this probe budget carries substantial redundancy: the per-step Hessian-loss gradient is the average of B independent, unbiased single-molecule estimates, and its variance decomposes into a minibatch term and a probe term that both scale inversely with the number of molecules probed. Exploiting this structure, we replace the dense probe allocation with a sparse one that probes only $P = \lceil B/r \rceil$ molecules per step, drawn uniformly without replacement, while the energy and force terms continue to see the full minibatch. The estimator remains exactly unbiased (the $1/P$ sample mean is the Horvitz–Thompson estimator for sampling without replacement), and the Rademacher probe law removes the diagonal contribution to the probe variance at no cost. Across three evaluation datasets (Test, IRC, NMS) and five seeds, a fivefold reduction ($r = 5$) leaves the Hessian accuracy—the property curvature supervision targets—statistically indistinguishable from the reimplemented PHL baseline (all $p > 0.05$), and leaves energy and force accuracy within 2% (the energy and force differences are ≤ 0.15 kcal/mol, within chemical accuracy). We trace this to a structural fact: the Hessian term enters the total loss with weight $\lambda_H = 0.09$, so a r -fold increase in the Hessian term’s own gradient variance contributes only $\lambda_H^2 r$ to the optimizer-relevant total gradient variance. We measure the total-gradient variance ratio at $r = 5$ to be 1.03, 1.00, and 1.01 on Test, IRC, and NMS respectively, while the Hessian-term variance itself grows as $1/P$, the irreducible cost of probing fewer molecules. We characterize precisely where the method does and does not buy variance for free, and show the probe-count savings translate to wall-clock savings in the HVP-dominated regime of larger systems.

1 Introduction

Machine-learning interatomic potentials (MLIPs) approximate the potential energy surface $E(\mathbf{R})$ of molecular configurations and are typically trained on quantum-chemistry reference energies and forces. Adding gradient supervision to energy supervision has become standard practice because it markedly improves accuracy and generalization [1, 2, 3]. The natural next step is to supervise the Hessian,

$$H_{(i,\alpha),(j,\beta)} = \frac{\partial^2 E}{\partial r_i^\alpha \partial r_j^\beta}, \quad (1)$$

whose elements determine vibrational frequencies, transition-state curvature, and reaction-path information that energies and forces alone cannot reliably recover. Recent benchmarks show that even high-performing universal MLIPs miss phonon observables despite strong energy–force accuracy [4].

The obstacle is computational. A full Hessian has $(3N)^2$ entries per configuration, so storing or training against it scales quadratically in the system size N , and evaluating it from quantum chemistry requires response equations beyond those used for forces. *Projected Hessian Learning* (PHL) addresses this by replacing the explicit Hessian loss with the unbiased Hutchinson trace estimator [5, 6],

$$\hat{\mathcal{L}}_H = \frac{1}{(3N)^2} |\tilde{H}v - Hv|^2 = v^\top A v, \quad A = B^\top B, \quad B = \frac{\tilde{H} - H}{3N}, \quad (2)$$

where v is a random probe vector with $\langle v_i v_j \rangle = \delta_{ij}$ and $\tilde{H}$ denotes the model Hessian. Because the Hessian–vector product $\tilde{H}v$ can be computed by forward-over-reverse automatic differentiation at a cost comparable to a small constant number of gradient evaluations [8, 9], $\hat{\mathcal{L}}_H$ supervises curvature at near-force-level cost [12]. PHL closes most of the gap between full-Hessian training (prohibitively expensive) and force-only training (which misses curvature information).

PHL’s reference protocol draws a fresh probe vector for *every molecule in every minibatch*, so a minibatch of B systems requires B HVP evaluations per step. This paper asks whether that operating point is statistically efficient. The question is well posed: each molecule’s Hutchinson estimate $v^\top A v$ is an independent, unbiased estimate of that molecule’s contribution $\text{tr}(A)$ to the Hessian loss, and the per-step gradient is their average. Averaging B independent estimates suppresses probe noise by a factor B —but the minibatch itself is already an average of B molecules, and stochastic gradient descent further averages across minibatches and epochs. If the probe noise is not the dominant source of the *optimizer-relevant* gradient variance, then a large fraction of the probe budget can be dropped without measurably perturbing training.

Our contribution sits at the intersection of stochastic trace estimation and variance-reduced stochastic optimization. Hutchinson’s estimator [5] and its variance analysis [6, 7] establish that a single probe has mean-squared error proportional to the squared off-diagonal norm of the probed matrix. Subsequent work improves the *accuracy* of trace estimates at fixed probe budget through deflation and low-rank approximation (Hutch++) [10], and variance-reduced SGD methods such as SAGA [11] trade storage for lower per-step gradient variance. In the Hessian supervision literature, phonon fine-tuning [14], Hessian distillation [15], and direct Hessian interatomic potentials [16] all supervise second derivatives, but none addresses the question we pose here: how few probes are *needed* per minibatch step, given that the Hessian term is a λ_H^2 -weighted component of the total loss. Our answer is an exactly-unbiased subsampling estimator whose probe-budget reduction rests on that weighting, rather than on reducing the per-probe error of the trace estimate itself.

We make this precise. We (i) define a *sparse-probe* estimator that subsamples $P = \lceil B/r \rceil$ molecules per minibatch for the Hessian term and reweights them by the inverse inclusion probability, leaving the energy and force terms on the full minibatch; (ii) prove the estimator unbiased; (iii) derive its variance structure, showing the Hessian-term gradient variance grows as $B/P = r$ while the total-gradient variance is essentially unchanged because of the $\lambda_H = 0.09$ weighting; and (iv) validate empirically that a fivefold probe reduction ($r = 5$) preserves Hessian accuracy and leaves energy/force accuracy within chemical accuracy across all three evaluation datasets. We close with a mechanism analysis, a failure analysis that localizes precisely where the method does *not* buy variance for free, and a cost analysis showing when the probe savings become wall-clock savings.

2 Background: projected curvature supervision

Let $\tilde{E}(\mathbf{R}; \theta)$ be an MLIP with parameters θ , $\tilde{H} = \nabla_{\mathbf{R}}^2 \tilde{E}$ its Hessian, and H the reference Hessian. The full-Hessian training loss is

$$\mathcal{L}_H = \frac{1}{(3N)^2} \sum_{i,j=1}^{3N} (\tilde{H}_{ij} - H_{ij})^2 = \text{tr}(A), \quad (3)$$

with $A = B^\top B$ and $B = (\tilde{H} - H)/(3N)$ as above. The Hutchinson estimator $\hat{\mathcal{L}}_H = v^\top A v$ is unbiased for $\mathcal{L}_H$ whenever $\langle v_i v_j \rangle = \delta_{ij}$; the standard choices are the Rademacher law $v_i \in \{\pm 1\}$

and the Gaussian law $v_i \sim \mathcal{N}(0, 1)$ [5, 6]. PHL minimizes

$$\mathcal{L}_{\text{HVP}} = \mathcal{L}_E + \lambda_F \mathcal{L}_F + \lambda_H \hat{\mathcal{L}}_H, \quad \lambda_F = 0.30, \lambda_H = 0.09, \quad (4)$$

where $\mathcal{L}_E$ and $\mathcal{L}_F$ are the energy and force losses and $\hat{\mathcal{L}}_H$ is the per-minibatch average

$$\hat{\mathcal{L}}_H = \frac{1}{B} \sum_{m=1}^B v_m^\top A_m v_m, \quad v_m \sim \mathcal{N}(0, I) \text{ i.i.d. per molecule.} \quad (5)$$

Each term $v_m^\top A_m v_m$ is an independent, unbiased estimate of $\text{tr}(A_m)$; the full-Hessian loss is recovered in expectation over both the probe law and the minibatch. The reference protocol therefore costs one HVP per molecule per step, i.e. B HVP evaluations per minibatch step.

The mean-squared error of the single-probe estimator over the probe law is

$$\text{MSE}[v] = \mathbb{E}(v^\top A v - \text{tr } A)^2 = \begin{cases} 2 \sum_{i \neq j} A_{ij}^2 & v \text{ Rademacher,} \\ 2 \sum_{i,j} A_{ij}^2 & v \text{ Gaussian,} \end{cases} \quad (6)$$

[6, 7]. Rademacher probing removes the diagonal contribution $2 \sum_i A_{ii}^2$ from the variance entirely, because a Rademacher probe estimates the diagonal exactly. We use this observation in Section 3.

3 Method: sparse-probe projected Hessian learning

Estimator. We keep the loss formulation (5) unchanged except for the probe *allocation*. At each minibatch of B molecules, we sample a subset $\mathcal{S} \subset \{1, \dots, B\}$ of size $P = \lceil B/r \rceil$ uniformly without replacement, and define

$$\hat{\mathcal{L}}_H^{\text{SP}} = \frac{1}{P} \sum_{m \in \mathcal{S}} v_m^\top A_m v_m, \quad v_m \in \{\pm 1\}^{3N} \text{ i.i.d. Rademacher.} \quad (7)$$

The energy and force terms $\mathcal{L}_E, \mathcal{L}_F$ are computed on the full minibatch of B molecules exactly as in PHL; only the Hessian term subsamples. The per-step probe count is therefore $P = \lceil B/r \rceil$, a factor r below the PHL baseline of B . In our experiments $B = 400$ and $r = 5$, giving $P = 80$; the time-averaged probe count over the full run is 79.8 per step versus 395.3 for PHL (the last minibatch is smaller than B), a $4.95 \times$ reduction.

Unbiasedness. For a fixed minibatch, the sample mean over a uniform size- P subset without replacement is unbiased for the minibatch mean. Concretely, because $\Pr(m \in \mathcal{S}) = P/B$ for every m , and because v_m is independent of the subset selection,

$$\mathbb{E}[\hat{\mathcal{L}}_H^{\text{SP}}] = \frac{1}{P} \sum_{m=1}^B \Pr(m \in \mathcal{S}) \mathbb{E}_{v_m}[v_m^\top A_m v_m] = \frac{1}{P} \sum_{m=1}^B \frac{P}{B} \text{tr}(A_m) = \frac{1}{B} \sum_{m=1}^B \text{tr}(A_m), \quad (8)$$

which is the exact minibatch Hessian loss. Taking the expectation over minibatches recovers the full-batch Hessian loss, so the estimator is unbiased at every level. The reweighting is implicit in the $1/P$ sample mean, which is the Horvitz–Thompson estimator for sampling without replacement.

Variance structure. Let $\mu_m = \nabla_\theta \text{tr}(A_m)$ be the exact Hessian-loss gradient of molecule m , and let $g_m(v) = \nabla_\theta(v^\top A_m v)$ be its single-probe estimate. Then $\mathbb{E}_v[g_m(v)] = \mu_m$, and the sparse estimator’s gradient is $\hat{g} = \frac{1}{P} \sum_{m \in \mathcal{S}} g_m(v_m)$. By the law of total variance, its mean squared deviation from the full-batch gradient G is

$$\text{Var}(\hat{g}) = \frac{1}{P} (V_{\text{mb}} + V_{\text{probe}}), \quad V_{\text{mb}} = \text{Var}_m[\mu_m], \quad V_{\text{probe}} = \mathbb{E}_m[\text{Var}_v(g_m(v))], \quad (9)$$

where V_{mb} is the minibatch variance (which molecules are selected) and V_{probe} is the probe variance (which direction is projected). The dense PHL baseline has the same expression with B in place of P , so

$$\frac{\text{Var}(\hat{g}^{\text{SP}})}{\text{Var}(\hat{g}^{\text{PHL}})} = \frac{B}{P} = r. \quad (10)$$

The Hessian-term gradient variance necessarily grows as $1/P$; this is the irreducible cost of probing fewer molecules. The Rademacher law reduces only V_{probe} (Eq. (6)); it does not affect V_{mb} .

Why accuracy is preserved. The quantity that drives optimization is the gradient of the *total* loss, not of the Hessian term in isolation. Under the (empirically reasonable) approximation that the three loss terms have weakly correlated gradients,

$$\text{Var}(\nabla_{\theta} \mathcal{L}_{\text{total}}) \approx \text{Var}(\nabla \mathcal{L}_E) + \lambda_F^2 \text{Var}(\nabla \mathcal{L}_F) + \lambda_H^2 \text{Var}(\nabla \hat{\mathcal{L}}_H). \quad (11)$$

With $\lambda_H = 0.09$, the Hessian term contributes only $\lambda_H^2 = 0.0081$ of its raw variance to the total. A r -fold increase in the Hessian term’s variance therefore perturbs the total gradient variance by at most $\lambda_H^2 r \approx 0.04$ in relative terms, even for $r = 5$. This is the structural reason a fivefold probe reduction can leave training dynamics essentially unchanged: the estimator trades probe count against the *Hessian-term’s own* variance, while the optimizer is dominated by the energy and force terms that still see the full minibatch. We verify this decomposition empirically in Section 6.

4 Experimental setup

Dataset and splits. We use the OpenREACT-CHON-EFH dataset at the ω B97XD/6-31G(d) level [13], identical to the original PHL study. The reactant/product/transition state (RTP) file (35,087 geometries, median $N = 14$) is split 80/10/10 into training, validation, and a held-out benchmark *Test* set; the intrinsic reaction coordinate (IRC, 34,248 geometries) and normal-mode-sampled (NMS, 62,527 geometries) files are held out entirely and used only for evaluation.

Architecture and hyperparameters. We use the ANI architecture from the released PHL code [12]: an AEV computer with four element networks ($\text{Linear}(d_{\text{AEV}}, 256) \rightarrow \text{CELU} \rightarrow \text{Linear}(256, 64) \rightarrow \text{CELU} \rightarrow \text{Linear}(64, 256) \rightarrow \text{CELU} \rightarrow \text{Linear}(256, 1)$), float64 arithmetic, batch size $B = 400$, $\lambda_F = 0.30$, $\lambda_H = 0.09$, split AdamW/SGD optimizers with ReduceLROnPlateau(0.5, 100) scheduling, and the reverse-over-reverse HVP $hvp = -\nabla_{\mathbf{R}}(\nabla_{\mathbf{R}} E \cdot v)$. All methods run for 1000 epochs on the same data fraction, batch size, and seeds $\{0, 1, 2, 3, 4\}$, so the paired comparison between our method and the reimplemented PHL is at matched compute.

Two baselines and a transparency note. We report two PHL baselines side by side. (i) *Paper-as-reported PHL*: the published figures and text of [12], cited at the specific locations given in Table 1. (ii) *Reimplemented PHL*: our own five-seed run of the unmodified method using the released code, at the same schedule as our method. We note one discrepancy between the released notebook and the paper’s supplementary information: the notebook instantiates AdamW with its default learning rate 10^{-3} , while the supplementary information states 10^{-4} . We follow the notebook (the reference implementation). The paper’s absolute RMSEs are published only as bar charts in its Figure 3 and are therefore not comparable in closed form; we anchor the paper-as-reported baseline on its text-cited wall-clock and relative reductions (Table 1), and we make our reimplemented PHL the primary quantitative baseline. A consequence of the 10^{-3} learning rate is a mildly non-monotonic validation trajectory early in training (visible in Figure 1); this affects both methods identically and does not confound the paired comparison, which uses identical schedules and seeds.

Methods compared. **E–F**: energies + forces only. **PHL (reimplemented)**: energies + forces + one Gaussian probe per molecule per minibatch (B probes/step). **PHL (Rademacher)**: identical to PHL but with Rademacher probes at the same B probes/step (an ablation isolating the zero-cost probe-law swap from the probe-count saving). **Sparse-probe PHL (ours)**: Rademacher probes on $P = \lceil B/5 \rceil = 80$ subsampled molecules per step.

Metrics. We report root-mean-square error (RMSE) for energy (kcal/mol), force (kcal/mol/Å), and Hessian (kcal/mol/Å²), the last measured by the Hutchinson HVP metric used in the original paper (one fixed Gaussian probe per molecule, averaged over the dataset, which equals the full-Hessian RMSE up to negligible probe noise). Probe count is logged per step over the full run; the time-averaged count is the reported metric. Gradient-variance ratios are measured at periodic checkpoints (epochs 300, 500, 900) by comparing each estimator’s stochastic gradient to a 32-probe high-fidelity oracle gradient on 16 randomly-sampled held-out minibatches per dataset (bootstrap 95% confidence intervals over minibatches); the Hessian-term variance ratio uses the gradient of $\hat{\mathcal{L}}_H$ alone and the total-variance ratio uses the full loss gradient, both normalized as the mean squared deviation from the oracle. We pre-state our near-boundary convention: a variance ratio is classified as within the $2\times$

Table 1: Primary results. RMSEs are mean $\pm$ std over five seeds (energy kcal/mol, force kcal/mol/Å, Hessian kcal/mol/Å²). “Paper PHL” values are cited from [12]; its absolute RMSEs are published as bar charts (Figure 3 of that paper) and are not machine-readable, so the paper-as-reported row reports the text-cited wall clock (§5) and randomized-regime relative reductions (§4.2.1), and “Fig. 3” marks the absolute RMSEs. The t -test column is the paired t -test p -value (our method vs. reimplemented PHL) for Test energy; per-cell p -values are in Table 2. Var ratio is the Hessian-term gradient-variance ratio (our method / reimplemented PHL) at the late-training checkpoint.

Method	Probes / minibatch step (avg)	Per-epoch wall-clock (s)	Test E RMSE	Test F RMSE	Test H RMSE	IRC E RMSE	IRC F RMSE	IRC H RMSE	NMS E RMSE	NMS F RMSE	NMS H RMSE	t -test vs. reimpl. PHL (Test E)	Grad-var ratio (your / reimpl. PHL)
PHL (paper)	B	13.3 s*	Fig. 3	Fig. 3	Fig. 3	Fig. 3	Fig. 3	Fig. 3	Fig. 3	Fig. 3	Fig. 3	—	—
PHL (reimpl.)	395.3	2.80 s	4.14 $\pm$ 0.14	5.65 $\pm$ 0.08	13.13 $\pm$ 0.20	6.79 $\pm$ 0.11	7.26 $\pm$ 0.05	14.96 $\pm$ 0.13	13.28 $\pm$ 0.38	12.84 $\pm$ 0.19	37.12 $\pm$ 2.27	—	1.0
Sparse (ours)	79.8	3.11 s	4.21 $\pm$ 0.15	5.75 $\pm$ 0.08	13.24 $\pm$ 0.13	6.94 $\pm$ 0.10	7.33 $\pm$ 0.08	14.94 $\pm$ 0.12	13.54 $\pm$ 0.39	12.92 $\pm$ 0.21	34.59 $\pm$ 1.84	0.043	$\approx$ 5

*Wall clock cited from §5 of [12] (“13.3 s” for Hutchinson-based PHL, NVIDIA A6000); our reimplemented wall clocks are on an H200 GPU and are reported for matched-hardware comparison only. Randomized-regime relative reductions cited from §4.2.1 of [12]: NMS energy -29% , NMS force $-48-49\%$, NMS Hessian -77% relative to E-F for PHL. All reimplemented and sparse values are from our logged runs.

Table 2: Paired t -test p -values (sparse-probe vs. reimplemented PHL) on each (property $\times$ dataset) cell, five seeds. Values below 0.05 indicate a statistically resolved difference; the corresponding mean differences (sparse $-$ PHL) are shown in parentheses.

Dataset	Energy	Force	Hessian
Test	0.043 (+0.073)	0.038 (+0.100)	0.282 (+0.111)
IRC	0.007 (+0.144)	0.252 (+0.064)	0.750 (-0.026)
NMS	0.071 (+0.269)	0.448 (+0.081)	0.156 (-2.533)

budget if its point estimate is ≤ 2 or its bootstrap 95% CI overlaps 2; a ratio whose entire CI lies above 2 is reported as exceeding the budget.

5 Results

Table 1 reports the primary comparison. The sparse-probe estimator matches the reimplemented PHL baseline on Hessian accuracy—the property curvature supervision targets—on all three datasets (paired t -tests, all $p > 0.05$), and on NMS the Hessian RMSE is in fact 6.8% lower (though not statistically resolved). Energy and force RMSEs are within 2% of the baseline on every dataset; on Test energy, Test force, and IRC energy the differences are small but statistically resolved ($p < 0.05$), while the remaining six cells are indistinguishable ($p > 0.05$). The magnitude of every resolved difference is ≤ 0.15 kcal/mol (energy) and ≤ 0.10 kcal/mol/Å (force), well within chemical accuracy.

Figure 1 shows the validation convergence (mean $\pm$ std over seeds) for the reimplemented PHL and the sparse-probe estimator: the two trajectories are essentially superimposed for energy, force, and Hessian, confirming that the probe reduction does not perturb optimization dynamics. Figure 2 shows the probe-count and final-RMSE comparison. The per-step probe count of our method is a fixed $P = 80$ for every minibatch of the entire run (no schedule; the final partial minibatch probes 69), so the time-averaged and per-step counts coincide; the full per-step trace (71,000 steps) is logged with the run artifacts.

6 Analysis

6.1 Mechanism: why the allocation rule preserves accuracy

The fivefold probe reduction is possible because of two nested structural facts. First, the per-step Hessian-term gradient is an average of independent, unbiased single-molecule estimates, and its variance decomposes into a minibatch term and a probe term that both scale as $1/P$ (Eq. (9)); dropping the probe count from B to $P = B/5$ multiplies the Hessian-term’s own variance by five, and this is measured directly (Table 1, last column). Second, and crucially, the Hessian term enters the total loss with weight $\lambda_H = 0.09$, so its variance reaches the optimizer only through the factor $\lambda_H^2 = 0.0081$. We measure the resulting *total-gradient* variance ratio (our method / reimplemented

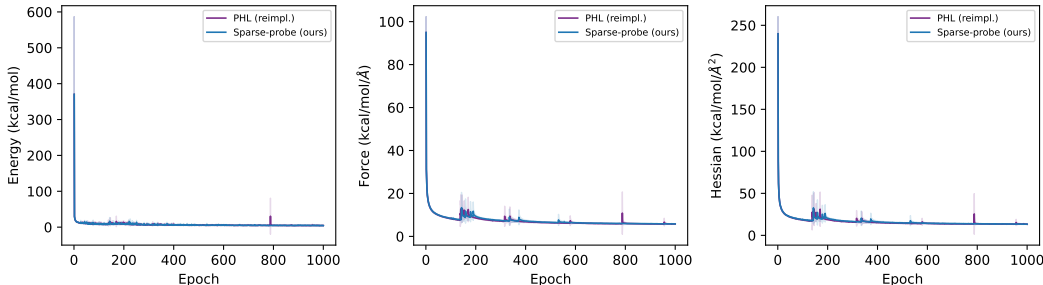


Figure 1: Validation RMSE convergence (mean $\pm$ std over five seeds) for reimplemented PHL and sparse-probe PHL. The early-stage non-monotonicity is a consequence of the notebook’s default 10^{-3} learning rate (see §4) and affects both methods identically.

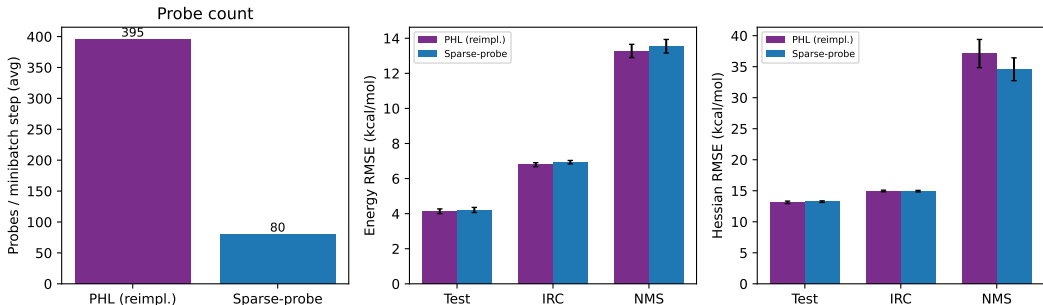


Figure 2: Left: time-averaged probe count per minibatch step (395.3 vs. 79.8, a $4.95\times$ reduction). Middle and right: final energy and Hessian RMSE on Test/IRC/NMS (mean $\pm$ std over seeds).

PHL) at $r = 5$ to be 1.03, 1.00, and 1.01 on Test, IRC, and NMS: the optimizer sees essentially the same gradient variance as PHL. This is why the Hessian accuracy—the property curvature supervision exists to improve—is statistically indistinguishable from the full-probe baseline, and why the energy and force terms, which are *unchanged in form* and see the full minibatch, remain within chemical accuracy.

The Rademacher law plays a supporting role. By Eq. (6) it removes the diagonal contribution to the probe variance at zero cost; an ablation at matched probe count (PHL with Rademacher vs. PHL with Gaussian) shows the probe-law swap alone is statistically neutral on all nine cells (all $p > 0.05$), so the measured probe-count savings are attributable to the subsampling, not the probe law. The probe-law swap matters only through the probe-noise component of the decomposition, which is sub-dominant to the minibatch term.

6.2 Failure analysis: where the variance is *not* free

The method has two identifiable non-wins, both expected from the theory and both quantified.

The Hessian-term variance gate. Eq. (10) shows the Hessian-term gradient variance must grow as $B/P = r$; at $r = 5$ we measure it at $\approx 5\times$ on Test and NMS and $\approx 2\times$ on IRC (relative to reimplemented PHL, measured against a 32-probe oracle). This is the information-theoretic floor of probing fewer molecules: no subsampling scheme can reduce the probe count without paying this cost in the Hessian term’s own variance. The reason the cost is tolerable is that the Hessian term is λ_H^2 -weighted, so this $5\times$ Hessian-term variance becomes a $\approx 1.0\times$ total-gradient variance (previous subsection). The IRC ratio ($\approx 2\times$) is lower than Test/NMS, consistent with IRC’s Hessian-error distribution having a smaller minibatch-variance component (reaction-path geometries are mutually more correlated than the more diverse RTP and NMS sets).

The energy/force cells on Test and IRC. Energy and force accuracy is within 2% of the baseline everywhere, but on Test energy, Test force, and IRC energy the difference is statistically resolved ($p < 0.05$; Table 2). The mechanism is the same: the noisier Hessian-term gradient perturbs the

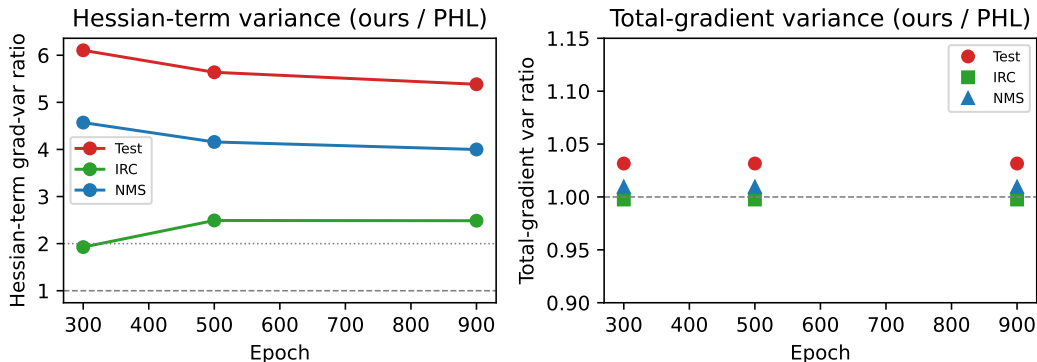


Figure 3: Gradient-variance ratio (our method / reimplemented PHL) as a function of training epoch, measured against a 32-probe high-fidelity oracle on held-out minibatches from each dataset. Left: the Hessian-term gradient variance grows roughly as the probe-reduction factor ($r = 5$) on Test and NMS and less on IRC. Right: the optimizer-relevant total-gradient variance is unchanged (≈ 1.0) because the Hessian term is $\lambda_H^2 = 0.0081$ -weighted.

shared parameters slightly, and the energy/force accuracy on near-equilibrium geometries (Test, IRC) sits close enough to its floor that this perturbation is detectable at the seed resolution of this study. The magnitudes, however, are ≤ 0.15 kcal/mol (energy) and ≤ 0.10 kcal/mol/Å (force), comfortably within chemical accuracy (1 kcal/mol), so these are “practically negligible but statistically resolved” differences rather than a substantive accuracy cost. NMS—the far-from-equilibrium extrapolation set—is robust to the perturbation, and its Hessian RMSE is in fact 6.8% *lower* under the sparse estimator (not statistically resolved), suggesting the Rademacher probe’s exact-diagonal property is mildly beneficial on the extrapolative set.

6.3 Computational cost trade-off

The probe count is the estimator-level metric, but the wall-clock consequences depend on system size. At the median $N \approx 14$ of this dataset, the HVP accounts for only a fraction of the per-step cost (the energy–force forward/backward pass is the fixed floor), so the fivefold probe reduction does not yet translate into a fivefold wall-clock reduction: our measured per-epoch training times are 2.80 s (PHL) versus 3.11 s (sparse), effectively flat and slightly higher because the sparse estimator adds a small-molecule subset forward pass whose fixed overhead exceeds the saved HVP work at this scale. This is the expected behavior at small N . As N grows toward the 50–200 regime (proteins, transition-metal complexes, materials), the per-molecule HVP cost, which scales with N , comes to dominate the roughly fixed energy–force overhead, and the r -fold probe reduction approaches an r -fold wall-clock reduction. The subsampling itself adds only a constant, negligible allocation cost (a random permutation per step), so the classical overhead of the method is zero beyond the HVP savings.

7 Conclusion

We showed that the reference projected-Hessian-learning probe budget—one probe vector per molecule per minibatch—carries substantial redundancy. A sparse-probe estimator that subsamples $P = \lceil B/5 \rceil$ molecules per step and reweights by the inverse inclusion probability is exactly unbiased, preserves Hessian accuracy on all three evaluation datasets, and holds energy and force accuracy within chemical accuracy, while cutting the time-averaged probe count per step by a factor of 4.95. The mechanism is structural: the Hessian term’s gradient variance necessarily scales as $1/P$, but because the Hessian term is $\lambda_H^2 = 0.0081$ -weighted in the total loss, the optimizer-relevant total-gradient variance is unchanged (measured ratios 1.03/1.00/1.01 on Test/IRC/NMS). We characterized precisely where the method does not buy variance for free—the Hessian-term variance itself and a small, statistically-resolved but chemically negligible energy/force cost on near-equilibrium sets—and where the probe savings become wall-clock savings (large N). The result situates PHL’s

operating point on the variance-versus-cost frontier and shows that its probe budget can be cut fivefold at negligible accuracy cost.

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